

Signals and Systems — Tutorial 6

System response and stability

Tutorial information

- It is expected that you attempt the prework questions before coming to the tutorial each week and submit your *handwritten* solutions to your tutor within the first 15 minutes of your tutorial.
- Not all questions are equally difficult. Difficult questions are marked with a ★.
- Questions marked **Extension** go beyond the assessable content for this course and are provided to give a deeper theoretical background and understanding.
- You may not be able to complete all questions within the tutorial time. It is *strongly* recommended that you attempt all questions in your own time and ask your tutor if you are stuck.

Pework questions

A: Impulse Response

Suppose the LTI systems $y_r(t)$, $y_b(t)$, $y_p(t)$ and $y_g(t)$ have impulse responses $h_r(t)$, $h_b(t)$, $h_p(t)$ and $h_g(t)$ shown in Figure 1. Identify which of the systems are BIBO stable.

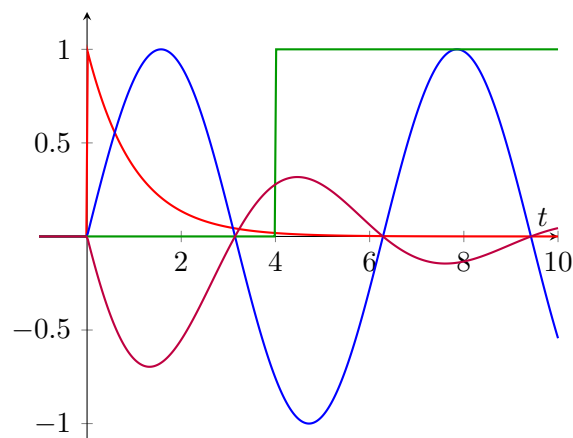


Figure 1: Impulse responses for the LTI systems y_r , y_b , y_p and y_g .

B: Stability

Let $\dot{y} = f(u(t), y(t))$ be a LTI system. Recall from Tutorial 1 the system has solutions of the form $y(t) = Ae^{zt} + B$ which we can plot in the complex plane.

Determine if the systems in the following plots are stable or unstable and if they are stable if they are critically/over-damped or under-damped.

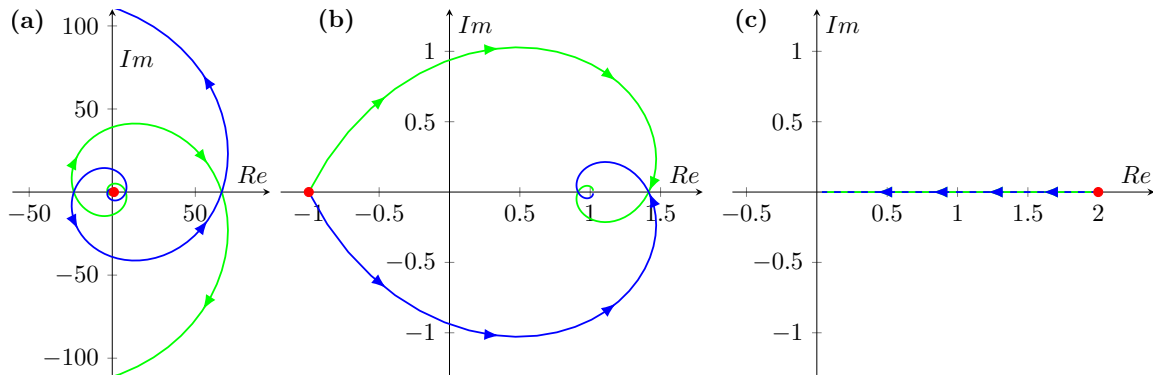


Figure 2: Plots of solutions $y(t) = Ae^{zt} + B$ where $A, B \in \mathbb{R}$, $z \in \mathbb{C}$ and $t \geq 0$. The blue line is when $z = x + jy$ and the green is when $z = x - jy$. The initial condition is shown by the red point.